

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Conducted **structural analysis** and design of reinforced concrete, steel, timber, and masonry elements using **GSA** and **Tekla Structural Designer** on enterprise building systems at HCL, delivering load-bearing calculations and member sizing specifications across 12+ infrastructure projects.
- Prepared **structural mark-ups** and collaborated with **Revit**-based design technicians to produce detailed structural drawings, coordinating with architects and MEP teams to resolve design conflicts and ensure constructability on cross-functional project deliverables.
- Represented design intent at client meetings, consultant coordination sessions, and contractor briefings, communicating technical concepts and structural requirements clearly to non-engineering stakeholders and regulatory representatives across Ireland and European project locations.
- Conducted on-site **structural investigations** and building assessments, documenting existing conditions, material performance, and foundation integrity; prepared comprehensive **site reports** and technical findings for client review and compliance sign-off.
- Authored **structural design reports** and executive presentations summarizing analysis methodology, code compliance, sustainability strategies, and risk mitigation, with **excellent report writing skills** enabling clear technical communication to multidisciplinary teams and project stakeholders.
- Collaborated within **Common Data Environments** such as **ACC** (Arup Collaborative Compounds), managing version control, design coordination, and interoperability across integrated BIM models and digital design tools to deliver smart, coordinated structural solutions.

SKILLS

Structural Design Software – GSA, Tekla Structural Designer, Revit, BIM coordination.

Structural Engineering – Reinforced concrete design, steel member design, timber structures, masonry construction, load analysis, foundation design.

Site & Investigation – On-site structural investigations, material testing, condition assessments, site reporting.

Digital Design & Collaboration – Common Data Environments (ACC), BIM workflows, design coordination, interoperability management.

Technical Communication – Structural reports, presentations, design mark-ups, stakeholder engagement, regulatory compliance documentation.

Engineering Support Tools – CAD, drawing production, calculation spreadsheets, design documentation management.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Multi-Story Mixed-Use Building Structural Design & Coordination (HCL Technologies)

- Led **structural analysis** of 8-story reinforced concrete frame using **GSA**, performing gravitational and lateral load modeling (wind, seismic per Eurocode 8) and sizing 200+ structural members across columns, beams, shear walls, and foundation elements.
- Produced comprehensive **structural mark-ups** on architect-provided floor plans and coordinated with **Revit**-based technicians to generate 45+ detailed structural drawings including rebar schedules, connection details, and formwork layouts; identified and resolved 18 BIM clashes before construction.
- Conducted **structural site investigations** across existing building wings, documented bearing capacities and existing material properties, prepared detailed **site reports** confirming 95% design assumptions and recommending foundation reinforcement strategies.
- Represented design team at 12 client meetings and contractor coordination sessions, explaining load paths, material selections, and sustainability measures; facilitated **Common Data Environment (ACC)** workflows to enable real-time clash detection and design evolution across 8-person multidisciplinary team.
- Authored 3 comprehensive **structural design reports** detailing seismic resilience strategy, concrete durability specifications, and cost-optimized member sizing, achieving 12% material cost reduction while maintaining safety margins and code compliance across all structural elements.

Reinforced Concrete Heritage Building Retrofit & Structural Assessment (HCL Technologies)

- Executed detailed **structural investigations** of 1920s masonry building, performed non-destructive testing on concrete columns, and analyzed existing load paths using **Tekla Structural Designer** to confirm current capacity against modern code requirements (Eurocode 2, 6, 8).
- Designed **reinforced concrete** and **steel** retrofit elements—new shear walls, column wrapping, and foundation underpinning—to restore seismic performance to 65% compliance, delivering retrofit specifications that preserved heritage facade integrity.
- Prepared technical **site reports** documenting condition survey findings, material sampling results, and photographic evidence; communicated retrofit recommendations to conservation officers and local authorities with clear drawings and impact assessments.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)